



**B.Sc. (General/Special) Degree
Programme
Faculty of Applied Sciences
University of Sri Jayewardenepura**

ICT 1022 Computer System Architecture

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Memory Systems



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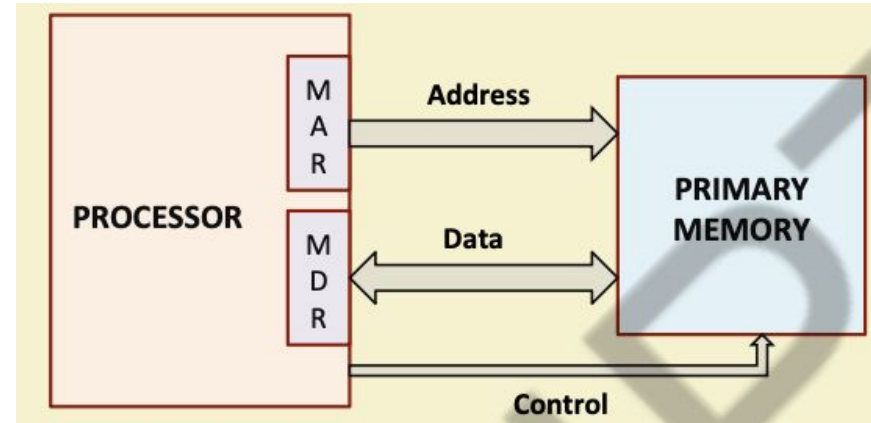
Introducion

- Memory is one of the most important functional units of a computer.
 - Used to store both instructions and data.
 - Stores as bits (0's and 1's), usually organized in terms of bytes.
- How are the data stored in memory accessed?
 - Every memory location has a unique address.
 - A memory is said to be byte addressable if every byte of data has a unique address.
 - Some memory systems are word addressable also (every addressed locations consists of mul8ple bytes, say, 32 bits or 4 bytes).



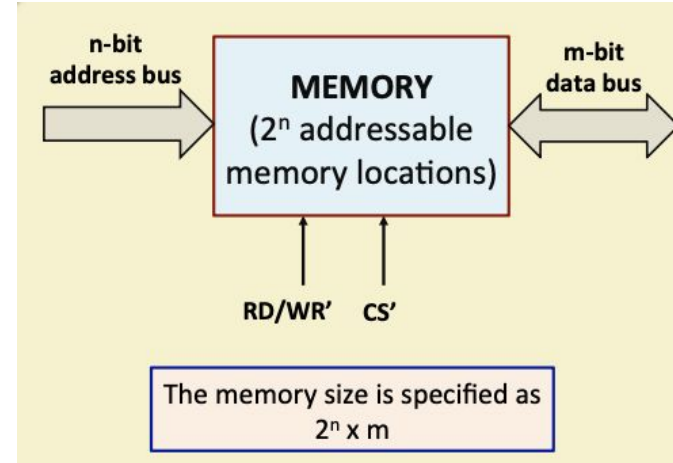
Connection between Processor and Memory

- Address bus provides the address of the memory location to be accessed.
- Data bus transfers the data read from memory, or data to be written into memory.
 - Bidirectional.
- Control bus provides various signals like READ, WRITE, etc



An Example Memory Module

- n address lines :: The maximum number of memory locations that can be accessed is 2^n .
- m data lines :: The number of bits stored in every addressable location is m .
- The RD/WR' control line selects the memory for reading or writing (1: read, 0: write).
- The chip select line (CS') when active (=0) will enable the chip; otherwise, the data bus is in the high impedance state.



Classification of Memory Systems

a) Volatile versus Non-volatile:

- A volatile memory system is one where the stored data is lost when the power is switched off.
 - Examples: CMOS static memory, CMOS dynamic memory.
 - Dynamic memory in addition requires periodic refreshing.
- A non-volatile memory system is one where the stored data is retained even when the power is switched off.
 - Examples: Read-only memory, Magnetic disk, CDROM/DVD, Flash memory, Resistive memory.



b) Random-access versus Direct/Sequential access:

- A memory is said to be random-access when the read/write time is independent of the memory location being accessed.
 - Examples: CMOS memory (RAM and ROM).
- A memory is said to be sequential access when the stored data can only be accessed sequentially in a particular order.
 - Examples: Magnetic tape, Punched paper tape.
- A memory is said to be direct or semi-random access when part of the access is sequential and part is random.
 - Example: Magnetic disk.
 - We can directly go to a track after which access will be sequential.



c) Read-only versus Random-access:

- Read-only Memory (ROM) is one where data once stored in permanent or semi-permanent.
 - Data written (programmed) during manufacture or in the laboratory.
 - Examples: ROM, PROM, EPROM, EEPROM.
- Random Access Memory (RAM) is one where data access time is the same independent of the location (address).
 - Used in main / cache memory systems.
 - Example: Static RAM (SRAM) → data once written are retained as long as power is on.
 - Example: Dynamic RAM (DRAM) → requires periodic refreshing even when power is on (data stored as charge on tiny capacitors).



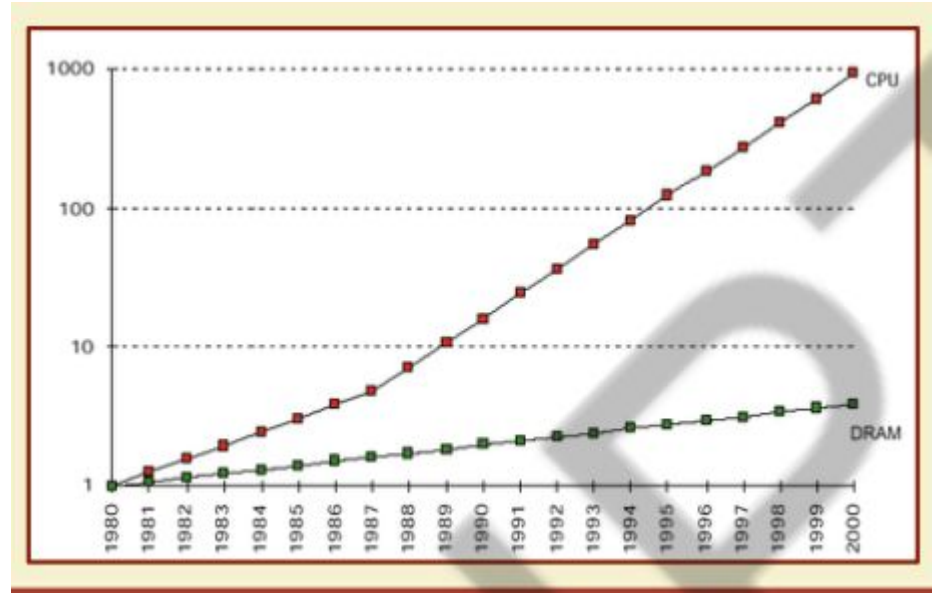
Access Time, Latency and Bandwidth

- Terminologies used to measure speed of the memory system.
 - a) Memory Access Time: Time between initiation of an operation (Read or Write) and completion of that operation.
 - b) Latency: Initial delay from the initiation of an operation to the time the first data is available.
 - c) Bandwidth: Maximum speed of data transfer in bytes per second.
- In modern memory organizations, every read request reads a block of words into some high-speed registers (LATENCY), from where data are supplied to the processor one by one (ACCESS TIME).



Design Issue of Memory System

- The most important issue is to bridge the processor-memory gap that has been widening with every passing year.
 - Advancements in memory technology are unable to cope with faster advancements in processor technology.



Design Issue of Memory System

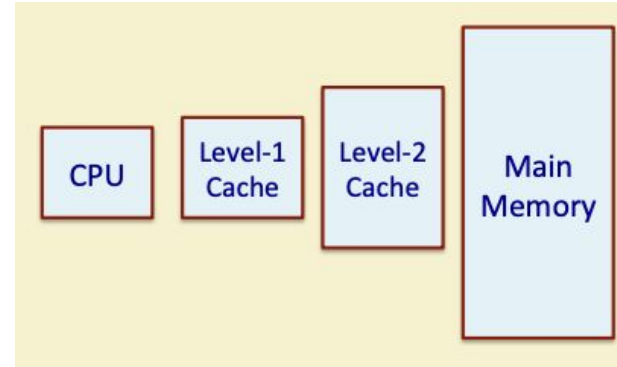
- Some important questions?
 - How to make the memory system work faster?
 - How to increase the data transfer rate between CPU and memory?
 - How to address the ever increasing storage needs of applications?
- Some possible solutions:
 - Cache Memory: to increase the effective speed of the memory system.
 - Virtual Memory: to increase the effective size of the memory system.



What is Cache Memory?

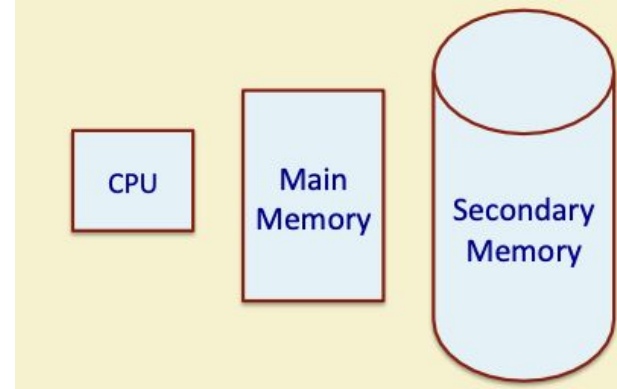
A fast memory (possibly organized in several levels) that sits between processor and main memory.

- Faster than main memory and relatively small.
- Frequently accessed data and instructions are stored here.
- Cache memory makes use of the fast SRAM technology.



What is Virtual Memory?

- Technique used by the operating system to provide an illusion of very large memory to the processor.
- Program and data are actually stored on secondary memory that is much larger.
- Transfer parts of program and data from secondary memory to main memory only when needed.



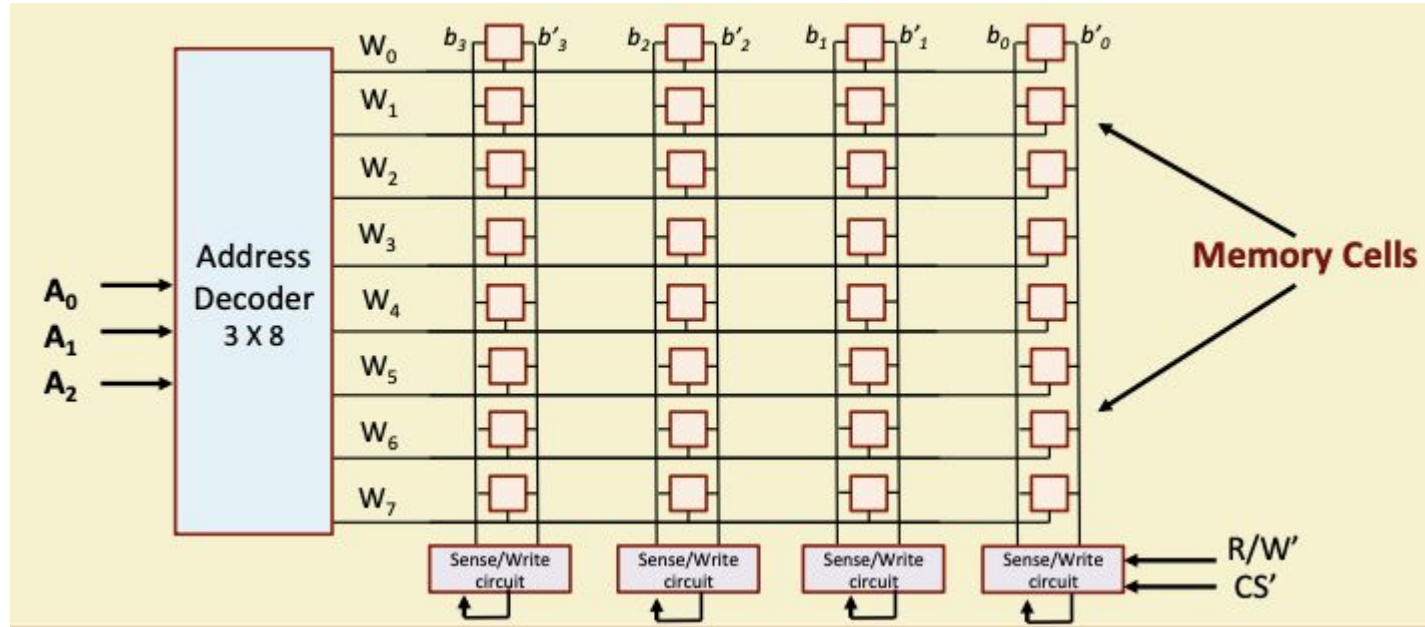
How a Memory Chip Looks Like?

Memory cells are organized in the form of an array.

- Every memory cell holds one bit of data.
- Present-day VLSI technology allows one to pack billions of bits per chip.
- A memory module used in computers typically contains several such chips.



Organization of Cells in an 8x4 Memory Chip



How a Memory Chip Looks Like?

- A 32-bit memory chip organized as 8 x 4 is shown.
- Every row of the cell array constitutes a memory word.
- A 3 x 8 decoder is required to access any one of the 8 rows.
- The rows of the cells are connected to the word lines.
- Individual cells are connected to two bit lines.
 - Bit b and its complement b' .
 - Required for reading and writing.
- Cells in each column are connected to a sense/write circuit by the two bit lines.
- Other than address and data lines, there are two control lines:
RW and CS' (Chip Select).



What About a 256 X 16 Memory?

- Here the total number of external connections are estimated as follows.
 - Address decoder size: 8×256
 - 8 external connections for address.
 - Data output : 16-bit
 - 16 external connections for data.
 - 2 external connections for R/W' and CS'.
 - 2 external connections for power supply and ground.
 - Total of $8+16+2+2=28$.



How a Memory Chip Looks Like?

External Connection Requirements

- The 8 x 4 memory requires the following external connections:
 - Address decoder of size: 3 x 8
 - 3 external connections for address.
 - Data output : 4-bit
 - 4 external connections for data.
 - 2 external connections for R/W' and CS'.
 - 2 external connections for power supply and ground.
 - Total of $3+4+2+2=11$.



- Programmers want unlimited amount of memory with very low latency.
- Fast memory technology is more expensive per bit than slower memory.
 - SRAM is more expensive than DRAM, DRAM is more expensive than disk.
- Possible solution?
 - Organize the memory system in several levels, called memory hierarchy.
 - Exploit temporal and spatial locality on computer programs.
 - Try to keep the commonly accessed segments of program / data in the faster memories.
 - Results in faster access times on the average.



Introduction

- Static RAM:
 - Very fast but expensive memory technology (requires 6 transistors / bit).
 - Packing density is limited.
- Dynamic RAM:
 - Significantly slower than SRAM, but much less expensive (1 transistor / bit).
 - Requires periodic refreshing.
- Flash memory:
 - Non-volatile memory technology that uses floating-gate MOS transistors.
 - Slower than DRAM, but higher packing density, and lower cost per bit.



Introduction

- Magnetic disk:
 - Provides large amount of storage, with very low cost per bit.
 - Much slower than DRAM, and also flash memory.
 - Requires mechanical moving parts, and uses magnetic recording technology.



Memory Hierarchy

The memory system is organized in several levels, using progressively faster technologies as we move towards the processor.

- The entire addressable memory space is available in the largest (but slowest) memory (typically, magnetic disk or flash storage).
- We incrementally add smaller (but faster) memories, each containing a subset of the data stored in the memory below it.
- We proceed in steps towards the processor.

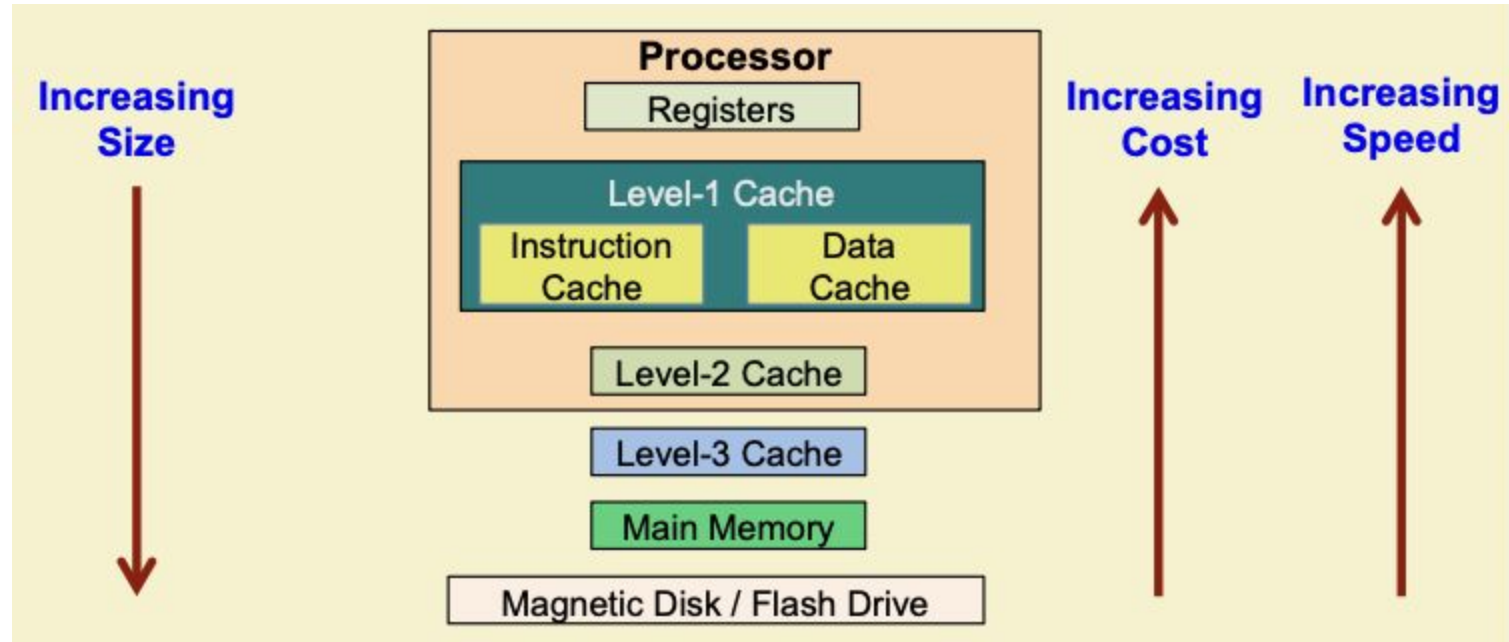


Memory Hierarchy

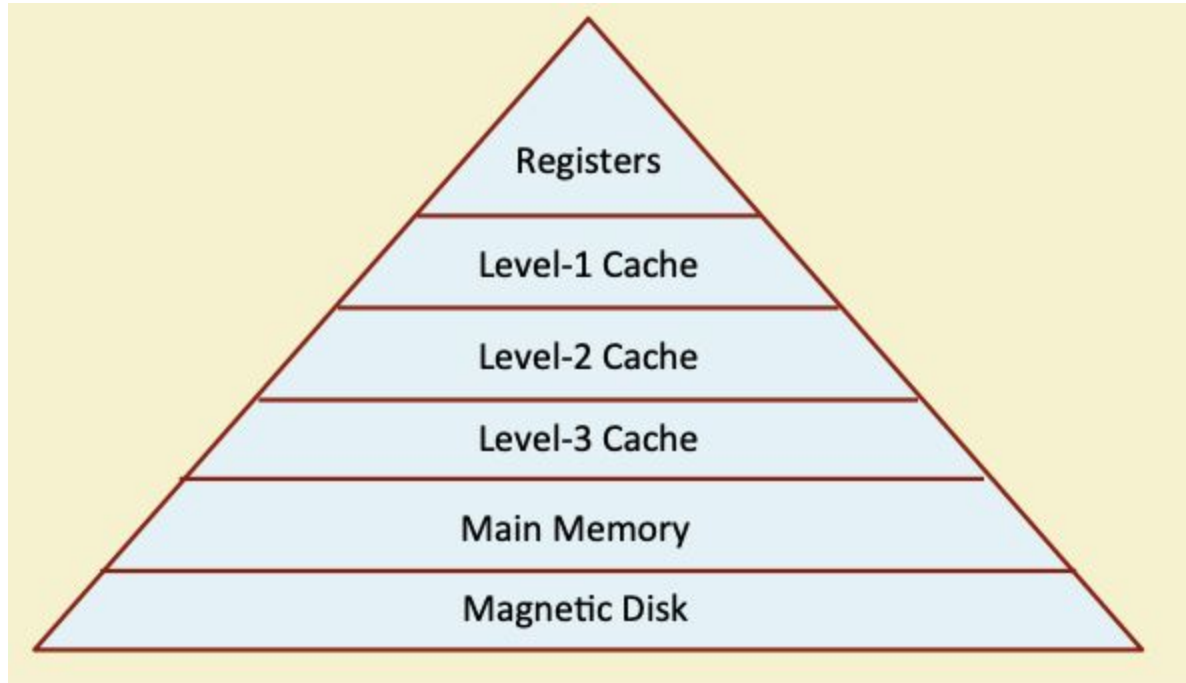
- Typical hierarchy (starting with closest to the processor):
 1. Processor registers
 2. Level-1 cache (typically divided into separate instruction and data cache)
 3. Level-2 cache
 4. Level-3 cache
 5. Main memory
 6. Secondary memory (magnetic disk / flash drive)
- As we move away from the processor:
 - Size increases
 - Cost decreases
 - Speed decreases



Memory Hierarchy



Memory Hierarchy



A Comparison

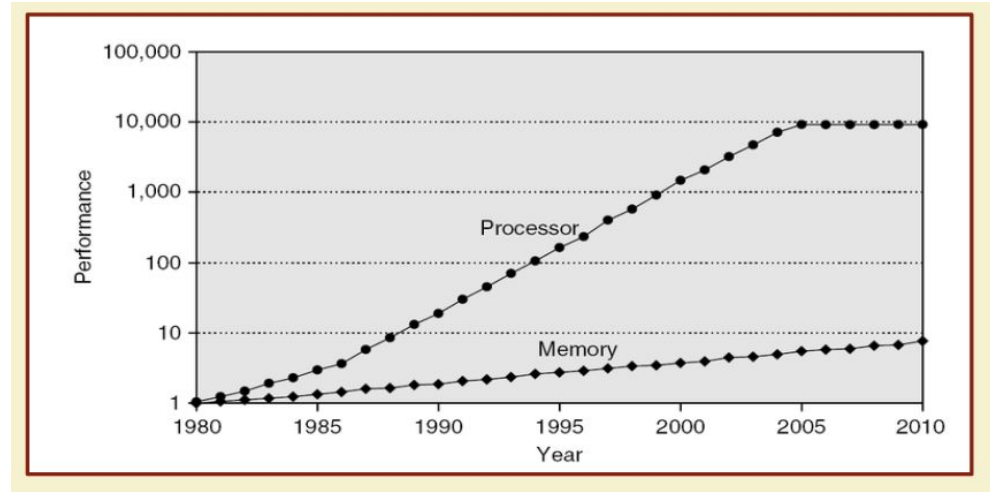
Level	Typical Access Time	Typical Capacity	Other Features
Register	300-500 ps	500-1000 B	On-chip
Level-1 cache	1-2 ns	16-64 KB	On-chip
Level-2 cache	5-20 ns	256 KB – 2 MB	On-chip
Level-3 cache	20-50 ns	1-32 MB	On or off chip
Main memory	50-100 ns	1-16 GB	
Magnetic disk	5-50 ms	100 GB – 16 TB	



Major Obstacle in Memory System Design

Processor is much faster than main memory.

- Has to spend much of the time waiting while instructions and data are being fetched from main memory.
- Memory speed cannot be increased beyond a certain point.



Impact of Processor / Memory Performance Gap

Year	CPU Clock	Clock Cycle	Memory Access	Minimum CPU Stall Cycles
1986	8 MHz	125 ns	190 ns	$190 / 125 - 1 = 0.5$
1989	33 MHz	30 ns	165 ns	$165 / 30 - 1 = 4.5$
1992	60 MHz	16.6 ns	120 ns	$120 / 16.6 - 1 = 6.2$
1996	200 MHz	5 ns	110 ns	$110 / 5 - 1 = 21.0$
1998	300 MHz	3.33 ns	100 ns	$100 / 3.33 - 1 = 29.0$
2000	1 GHz	1 ns	90 ns	$90 / 1 - 1 = 89.0$
2002	2 GHz	0.5 ns	80 ns	$80 / 0.5 - 1 = 159.0$
2004	3 GHz	0.33 ns	60 ns	$60 / 0.33 - 1 = 179.0$

Ideal memory access
time = 1 CPU cycle

Real memory access
time $\gg$ 1 CPU cycle



Impact of Processor / Memory Performance Gap

- Memory Latency Reduction Techniques:
 - Faster DRAM cells (depends on VLSI technology)
 - Wider memory bus width (fewer memory accesses needed)
 - Multiple memory banks
 - Integration of memory controller with processor
 - New emerging RAM technologies
- Memory Latency Hiding Techniques
 - Memory hierarchy (using SRAM-based cache memories)
 - Pre-fetching instructions and/or data from memory before they are actually needed (used to hide long memory access latency)



Impact of Processor / Memory Performance Gap

- Programs tend to reuse data and instructions they have used recently.
 - Rule of thumb: 90% of the total execution time of a program is spent in only 10% of the code (also called 90/10 rule).
 - Reason: nested loops in a program, few procedures calling each other repeatedly, arrays of data items being accessed sequentially, etc.
- Basic idea to exploit this rule:
 - Based on a program's recent past, we can predict with a reasonable accuracy what instructions and data will be accessed in the near future.

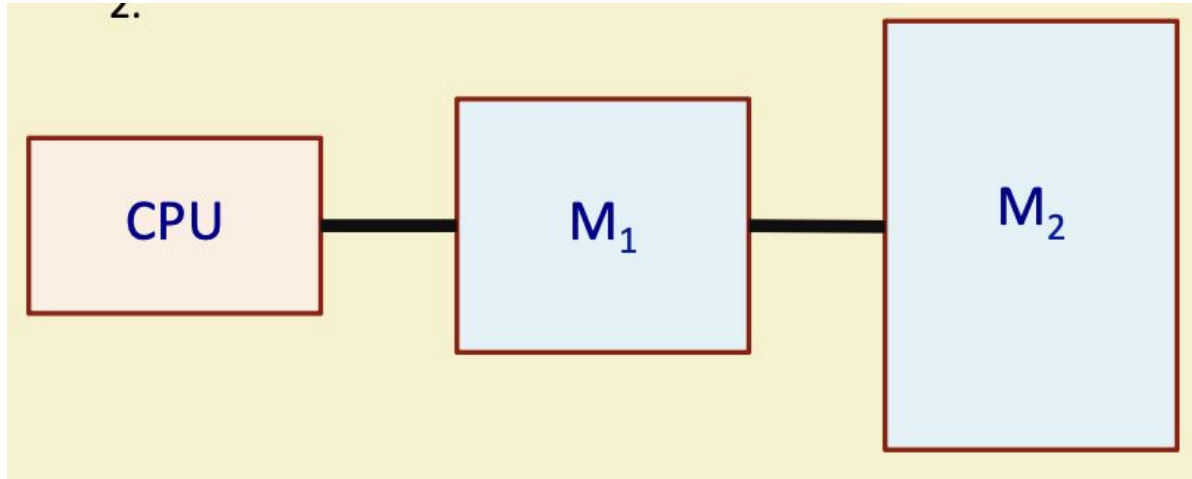


- The 90/10 rule has two dimensions:
 - a) Temporal Locality (locality in Time)
 - If an item is referenced in memory, it will tend to be referenced again soon.
 - b) Spatial locality (locality in space)
 - If an item is referenced in memory, nearby items will tend to be referenced soon.



Performance of Memory Hierarchy

We first consider a 2-level hierarchy consisting of two levels of memory, say, M_1 and M_2 .



Cost:

- Let c_i denote the cost per bit of memory M_i , and S_i denote the storage capacity in bits of M_i .
- The average cost per bit of the memory hierarchy is given by:

$$\text{Cost } c = \frac{c_1 S_1 + c_2 S_2}{S_1 + S_2}$$

- In order to have $c \rightarrow c_2$, we must ensure that $S_1 \ll S_2$.



Hit RaHo / Hit Rate:

- The hit ratio H is defined as the probability that a logical address generated by the CPU refers to information stored in M_1 .
- We can determine H experimentally as follows:
 - A set of representative programs is executed or simulated.
 - The number of references to M_1 and M_2 , denoted by N_1 and N_2 respectively, are recorded.

$$H = \frac{N_1}{N_1 + N_2}$$

- The quantity $(1 - H)$ is called the miss ratio.



Access Time:

- Let t_{A_1} and t_{A_2} denote the access times of M_1 and M_2 respectively, relative to the CPU.
- The average time required by the CPU to access a word in memory can be expressed as:

$$t_A = H.t_{A_1} + (1 - H).t_{MISS}$$

where t_{MISS} denotes the time required to handle the miss, called miss penalty.



The miss penalty t_{MISS} can be estimated in various ways:

- a) The simplest approach is to set $T_{\text{MISS}} = t_{A2}$, that is, when there is a miss the data is accessed directly from M_2 .
- b) A request for a word not in M_1 typically causes a block containing the requested word to be transferred from M_2 to M_1 . After completion of the block transfer, the word can be accessed in M_1 .

If t_B denotes the block transfer time, we can write

$$t_{\text{MISS}} = t_B + t_{A1} \text{ [since } t_B \gg t_{A1}, t_{A2} \approx t_B]$$

$$\text{Thus, } t_A = H.t_{A1} + (1 - H).(t_B + t_{A1})$$

- c) If t_{HIT} denotes the time required to check whether there is a hit, we can write

$$t_{\text{MISS}} = t_{\text{HIT}} + t_B + t_{A1}$$



Performance of Memory Hierarchy

Efficiency:

- Let $r = t_{A2} / t_{A1}$ denote the access time ratio of the two levels of memory.
- We define the access efficiency as $e = t_{A1} / t_A$, which is the factor by which t_A differs from its minimum possible value.

$$\begin{aligned} \text{Efficiency } e &= \frac{t_{A1}}{H \cdot t_{A1} + (1 - H) \cdot t_{A2}} \\ &= \frac{1}{H + (1 - H) \cdot r} \end{aligned}$$



Performance of Memory Hierarchy

Speedup:

- The speedup gained by using the memory hierarchy is defined as $S = t_{A2} / t_A$.
- We can write:

$$\begin{aligned} S &= \frac{t_{A2}}{H \cdot t_{A1} + (1 - H) \cdot t_{A2}} \\ &= \frac{1}{H / r + (1 - H)} \end{aligned}$$

- The same result follows from Amadahl's law.



Some Common Terminologies Used

- Block: The smallest unit of information transferred between two levels.
- Hit Rate: The fraction of memory accesses found in the upper level.
- Hit Time: Time to access the upper level
 - Upper level access Time + Time to determine hit/miss
- Miss: Data item needs to be retrieved from a block in the lower level.
- Miss Rate: The fraction of memory accesses not found in the upper level.
- Miss Penalty: Overhead whenever a miss occurs.
 - Time to replace a block in the upper level + Time to transfer the missed block



Thank
you!

dreamstime

